Regions and Availability Zones

spread across zones, plan for regions

Watch the 4-min video

In one line: a **region** is a geographic area of data centres; inside it, **availability zones** are independent sites a few kilometres apart, close enough for synchronous replication; deploying across zones survives a data-centre failure, but not a region outage, a bad global deploy or corrupted data, which need multi-region designs, safe rollouts and backups.

1 Same outage, different outcome

A power failure takes out one data centre in a cloud region. Company A, running on VMs in a single zone, is offline for six hours. Company B, in the same region, doesn't notice: its app and database were spread across three availability zones and traffic simply moved. The difference was a deployment decision, not luck.

2 Regions, zones and edge

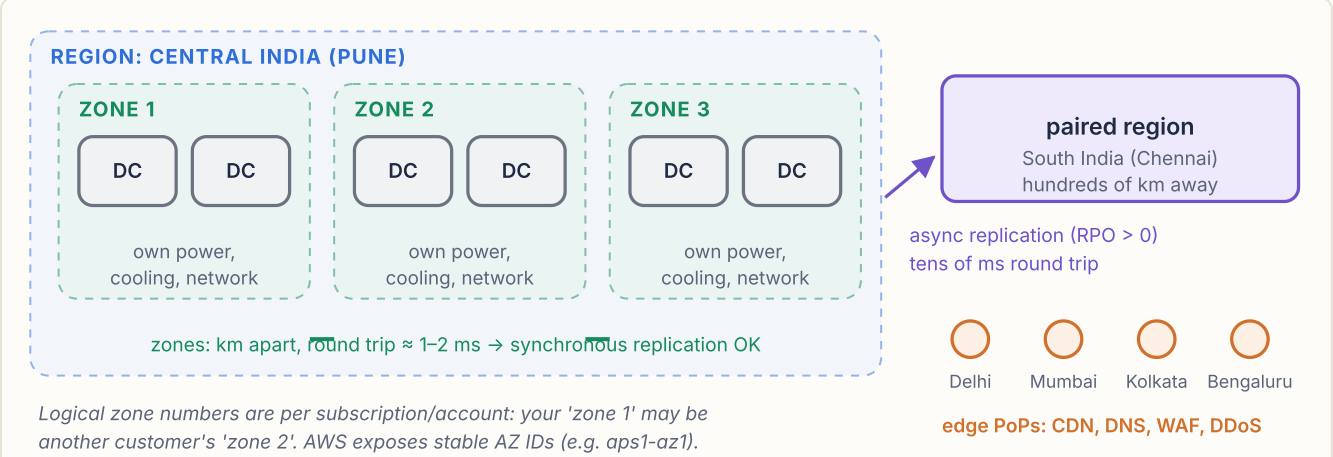


Figure 1. A region contains at least three zones (where zones are offered), each with independent facilities; edge locations sit much closer to users.

A **region** is a set of data centres in one metro or geography: Central India (Pune), South India (Chennai) and West India (Mumbai) on Azure; `ap-south-1` (Mumbai) on AWS. Regions are isolated from each other, so a disaster in one rarely affects another. Choose a region for:

- **Latency** to your users (roughly 1 ms of round trip per 100 km of fibre, plus routing).
- **Data residency and regulation**: e.g. the RBI's 2018 directive requires payment-system data in India to be stored only in India; GDPR constrains transfers out of the EU (lesson 46).
- **Service and capacity availability**: new services, GPU types and zone support arrive region by region.
- **Price**, which differs between regions, and a **paired / secondary region** for disaster recovery.

An **availability zone** is one or more data centres with independent power, cooling and networking. Zones in a region are far enough apart that a fire, flood or power failure hits only one, but connected by high-bandwidth, low-latency links: Microsoft describes round-trip latency between zones of under about 2 ms; AWS places its AZs within about 100 km of each other. That latency is low enough for **synchronous**

replication (RPO = 0) on every commit. Note that cross-zone traffic is not always free: AWS, for example, bills data transfer between AZs.

3 Zonal, zone-redundant, regional, global

Flavour	Behaviour	Examples	Your job
Zonal	pinned to one zone; fails with it	a VM, a managed disk, a zonal IP	deploy several, one per zone, and balance
Zone-redundant	provider spreads it across zones automatically	ZRS storage, zone-redundant Azure SQL, Standard LB, multi-AZ RDS	turn it on (often at creation only)
Regional	runs in the region; provider decides placement	many PaaS control planes, queues	check its documented zone resilience
Global	spans regions	DNS, Front Door / CloudFront, Entra ID / IAM	remember it is a shared dependency

SLAs reflect this. For Azure VMs, two or more instances across zones carry a 99.99% connectivity SLA, versus 99.95% in an availability set within one data centre and 99.9% for a single VM with premium storage. Stateful zonal pieces are the usual trap: a pod whose persistent disk is zonal can't restart in another zone unless the disk is zone-redundant or the data is replicated at the application level (lesson 59).

4 What multi-AZ does and doesn't buy you

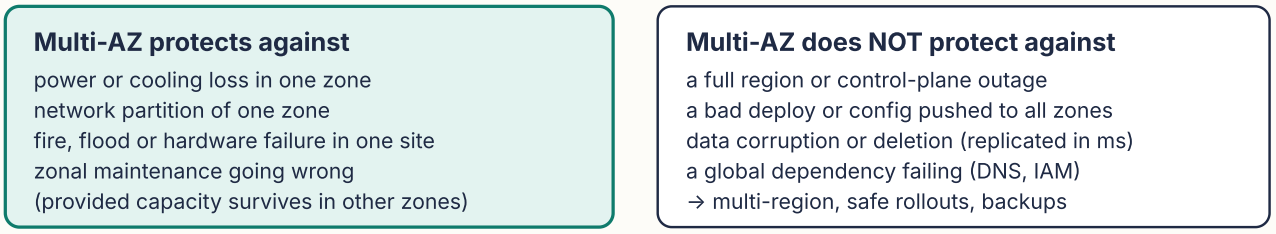


Figure 2. Zones remove the data centre as a single point of failure. Other failure modes need other defences.

Static stability matters as much as spreading: if losing a zone forces you to launch new capacity during the incident, you depend on the control plane exactly when it is busiest. Pre-provision so the survivors can absorb the load: with three zones, run each at no more than about two-thirds of its capacity (in total, ~150% of peak), so two zones can carry 100%. Region-level failures do happen, including control-plane outages such as AWS us-east-1 in December 2021, which is why critical systems add a second region (active-passive with asynchronous replication, or active-active) plus backups that a bad write can't reach.

5 Edge locations

Edge points of presence are small sites in hundreds of cities, much closer to users than regions. They host CDN caches, DNS resolvers, TLS termination, WAF and DDoS protection, and lightweight edge functions (lesson 69). Pushing static content and simple logic to the edge cuts latency for distant users and absorbs attacks, while the data and core logic stay in the region. Providers also offer in-between options such as AWS Local Zones and Azure Extended Zones for latency-sensitive workloads in specific metros.

WORKED EXAMPLE: THREE ZONES IN CENTRAL INDIA

The app runs on an AKS cluster whose node pool is created with zones 1, 2 and 3; pods use `topologySpreadConstraints` on `topology.kubernetes.io/zone` so 2 pods land in each zone. A zone-redundant Standard Load Balancer fronts them. Azure SQL runs in a zone-redundant configuration with synchronous replicas in other zones. Each zone has spare capacity for half the peak load.

Zone 2 loses power. Its nodes, 2 pods and a database replica disappear. Load-balancer health probes fail within seconds and traffic goes to zones 1 and 3; committed transactions are safe because replication was synchronous, and if the primary had been in zone 2, a replica would be promoted automatically. The remaining 4 pods absorb the load without waiting for autoscaling; Kubernetes reschedules the lost pods onto surviving zones. Users see a few failed requests that clients retry.

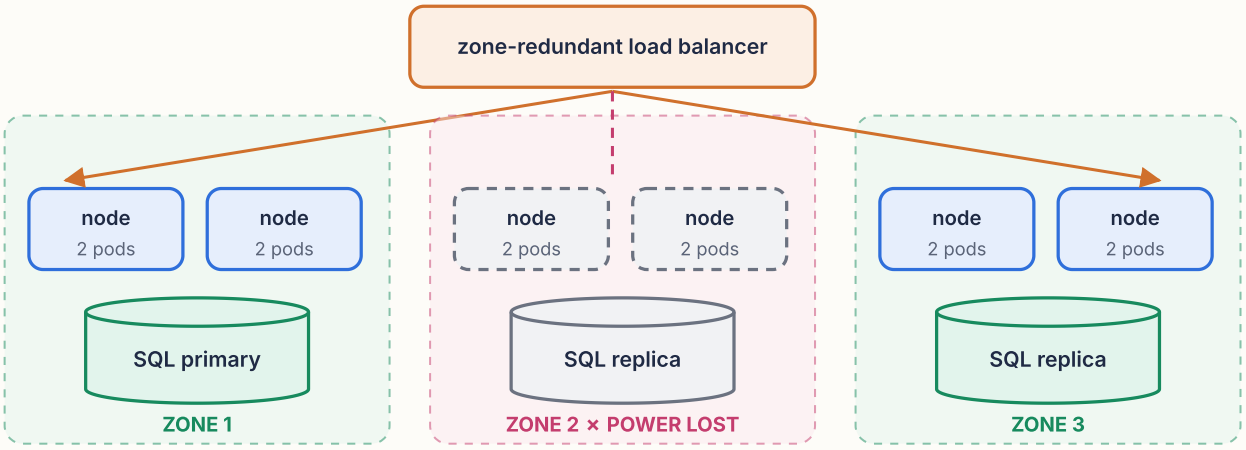


Figure 3. The worked example during the zone 2 failure: the load balancer stops sending traffic there and the other zones carry the load.

TRADE-OFFS

- **Multi-AZ**: survives a data-centre failure for modest cost, but adds cross-zone latency and (on some clouds) transfer fees.
- **Multi-region**: survives a region failure, but brings async replication (RPO > 0), failover complexity and roughly double cost.
- **Spare capacity per zone**: static stability costs money every day, but makes zone loss a non-event.
- **Region choice**: closest-to-users vs where the services, capacity and compliance rules are.

COMMON MISTAKES

- Running 'highly available' replicas that all landed in one zone.
- Making the app zone-redundant but leaving a zonal database, cache or disk as a single point of failure.
- Assuming multi-AZ protects against bad deploys or accidental deletes.
- Relying on autoscaling during a zone outage instead of pre-provisioned headroom.

6 Key terms

Region: geographic area with several data centres

Availability zone: independent site(s) in a region, ~1–2 ms apart

Zone-redundant: service spread across zones by the provider

Paired / secondary region: distant region for disaster recovery

Static stability: survive a failure without new provisioning

Edge PoP: small site near users: CDN, DNS, WAF

7 Interview questions

Q1. What failures does a multi-AZ deployment protect against? BASIC

Failures of a single data centre or zone, like power, cooling, networking or hardware, because zones are independent but close enough for synchronous replication.

Q2. What doesn't it protect against? BASIC

A full region outage, bad global deployments or replicated data corruption, which need multi-region designs, safe rollouts and backups.

Q3. How do you choose a region? BASIC

Latency to users, data-residency and compliance rules, availability of the services and capacity you need, price, and a suitable second region for DR.

Q4. You run 3 zones at 80% utilisation each. What happens when one fails? SCENARIO

The other two would need 120% each, so they overload until autoscaling adds capacity, if it can; keep each zone at ≤ ~66% for static stability.

Q5. Why might synchronous replication across regions be a bad idea? SCENARIO

Every commit waits for a round trip of tens of milliseconds or more, hurting latency and availability; regions usually replicate asynchronously and accept RPO > 0.

Q6. Your Kubernetes pods are spread across zones but the app still went down with zone 1. Why? TRICKY

A zonal dependency: a persistent disk, a single-zone database or cache, or a NAT/egress component pinned to zone 1.

Q7. Is 'zone 1' the same physical zone for every customer? TRICKY

Not necessarily: providers map logical zone names per subscription or account, so coordinate across accounts with AZ IDs or the provider's zone mapping.

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